

In Maryland

taxes 1.6 per person

$$= 1.6 \times \text{people}$$

income/GDP = 0.7 per person

pollution = 0.3 per person
 $0.3 \times \text{people}$

In Virginia

taxes = $1.4 \times \text{people}$

income = 0.6 / person

pollution = 0.5 / person

Total taxes = taxes in MD
+ taxes in VA

$$= 1.6 \cdot M + 1.4 \cdot V$$

Total income = income in MD
+ income in VA

$$= 0.7M + 0.6V$$

total poll = $0.3M + 0.5V$

From in Col

MD

VA

To in Taxes 1.6 1.4

Row Income 0.7 0.6

Pollution 0.3 0.5

Matrix

$$\begin{pmatrix} 1.6 & 1.4 \\ 0.7 & 0.6 \\ 0.3 & 0.5 \end{pmatrix}$$

in 1790 $M = 319700$
 $V = 691700$

$$\begin{pmatrix} \text{Taxes} \\ \text{Income} \\ \text{pollution} \end{pmatrix} = \begin{pmatrix} 1.6 & 1.4 \\ 0.7 & 0.6 \\ 0.3 & 0.5 \end{pmatrix} \begin{pmatrix} M \\ V \end{pmatrix} = \begin{pmatrix} 1.6M + 1.4V \\ 0.7M + 0.6V \\ 0.3M + 0.5V \end{pmatrix}$$

data is 4×2

tip matrix is 3×2

tip $\cdot$ data not allowed

A B $\leftarrow$
 $\uparrow$
cols of A = rows of B

~~A = transpose (tip)~~

B = transpose (data)

~~$\begin{bmatrix} 2.1 & 0.7 & 0.3 \\ 1.4 & 0.6 & 0.5 \end{bmatrix}$~~

2×3

tip $\cdot$ B is allowed

$3 \times 2 \quad 2 \times 4 = 3 \times 4$
 $\leftarrow$

$$B = \begin{matrix} & 1790 & 1800 & \overset{\text{From}}{1810} & 1820 \\ \begin{matrix} \text{To MD} \\ \text{To VA} \end{matrix} & \left(\begin{matrix} \\ \\ \end{matrix} \right) \end{matrix}$$

tip. $B = \begin{matrix} \text{From is "From} \\ \text{of B "} \end{matrix}$

$$\begin{matrix} \text{Rows} \\ \text{are} \\ \text{To of A} \end{matrix} \begin{matrix} \text{taxes} \\ \text{inc} \\ \text{pol.} \end{matrix} \begin{matrix} 1790 & 1800 & 1810 & 1820 \\ & & & 591370 \end{matrix}$$

↑
total pollution
in 1820